

Sadeeq Adeyemi

Perception Engineer, Multi-Object Tracking and Sensor Fusion

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SUMMARY

Perception engineer who builds and measures multi-object tracking and sensor fusion for autonomous vehicles and robots. Works in Python, C++ and PyTorch across camera, lidar and radar, combining Kalman filtering and data association with deep learning for detection and segmentation. Raised identity retention through occlusion from **0% to 89%** in simulation, then mined **10,871 occlusion events** from 1,256 real NVIDIA driving clips. Writes tested, reproducible code and works daily with AI coding agents.

SKILLS

Core stack: Python, C++, PyTorch, ROS 2, OpenCV, NumPy, Git, Docker

Tracking and fusion: multi-object tracking, sensor fusion, EKF and UKF state estimation, JPDA and MHT data association

Perception: computer vision, deep learning, 3D object detection and tracking, instance segmentation, re-identification, SAM 3, YOLOv8

Sensors and data: camera, lidar point clouds, radar, event cameras, NVIDIA PhysicalAI-AV, KITTI, BDD100K, Argoverse 2

Practice: HOTA, IDF1 and MOTA metrics, Monte Carlo evaluation, automated testing, MATLAB, AWS, AI coding agents

EXPERIENCE

Robotics Research Assistant, University of Maine Robotics Laboratory 2025 - Present

- Build mobile robot platforms end to end, from sensor integration and wiring to ROS 2 nodes and point cloud perception.
- Implemented EKF localization that fuses IMU, wheel odometry and lidar, with costmap planning and obstacle avoidance.

Materials Engineer, R.W. Gillespie, Maine. Tested construction materials to ASTM standards. 2024

Machine Learning Engineer and Structural Steel Fabricator, NewTech Steel, Nigeria

2018 - 2023

- Deployed a computer vision model for real-time fatigue and defect detection in steel, and cut test time by **50%** through automation.
- Led structural steel fabrication and erection for factory and auditorium projects.

PROJECTS

Occlusion-Aware Multi-Object Tracking 2026

- Built a tracker with EKF state estimation and Hungarian data association that models when each target is hidden from the sensor.
- Kept the correct identity through 1 to 5 second occlusions in **89% of cases, up from 0%**, across 20 Monte Carlo seeds in simulation.
- Cut identity switches per run from **75.6 to 6.6** by fusing camera, lidar and a second camera viewpoint.
- Validated against MathWorks GNN, JPDA and TOMHT trackers. Python, MATLAB and JavaScript implementations agree exactly.
- Runs in real time at **1.6 ms per frame**. Ported the companion navigation EKF to C++17 at 0.1 microseconds per step.
- Extended it with appearance re-identification, lidar instance segmentation at 0.94 mask IoU and an event camera model.
- Shipped a live 3D browser lab with 46 automated tests. Code at github.com/sadeeqgandalf/lunar-occlusion-tracking

Occlusion Mining on Real Driving Data, NVIDIA PhysicalAI-AV Dataset

2026

- Mined **10,871 pedestrian occlusion events** from 1,256 driving clips and 9.6 million 3D boxes with a ray-cast line-of-sight test.
- Projected lidar into the fisheye camera from calibration data, corrected motion distortion and verified time synchronization to 10 ms.
- Built a lidar depth-buffer visibility check and 3D review tooling for point clouds, blind zones and tracks.
- Pulled single clips from a 133 TB dataset with HTTP range requests, cutting each download from 2 GB to 10 MB.

Closed-Loop Autonomous Driving Stack

2026

- Integrated EKF localization, open-vocabulary perception across camera, lidar and radar, tracking, planning and LQR control.
- Passed **100 of 100** Monte Carlo runs with zero collisions. Without occlusion reasoning, 5 of 10 hidden-pedestrian runs collide.

Additional Perception Work

2025 - 2026

- Benchmarked SAM 3 against SAM 3.1 on KITTI-MOTS and MOTS20 with HOTA on an A100 GPU: IDF1 **84.8 vs 68.3** on MOTS20.
- Built a dashcam pipeline with YOLOv8 detection, tracking, lane detection and time to collision, covered by 24 regression tests.
- Built an FMCW radar chain in MATLAB from raw samples to CFAR detections, checked against the Cramer-Rao bound.

EDUCATION

MS, Data Science and Engineering, University of Maine

2025 - Present

MS, Structural Engineering, University of Maine

2023 - 2025

MSc, Project Management, University of Portsmouth, United Kingdom

2023

BEng, Civil Engineering, Federal University of Technology Akure, Nigeria

2019

Certifications: AWS Certified Machine Learning Specialty, AWS Certified Solutions Architect Associate